IMMERSIVE FUTURES

Assignment 02 – Web-Based Augmented Reality (AR) Experience Development

Training Program: Immersive Futures
Module: Web-Based Augmented Reality
Assignment Type: Individual Practical Assignment
Total Marks: 50

Submission Deadline: 26 October 2026

1. Assignment Introduction

In this assignment, you will create your own Web-Based Augmented Reality (WebAR) experience using an online AR development platform.

Your project should allow users to view a 3D object or scene in their real-world surroundings using a smartphone. The experience must run through a web browser without requiring users to download a separate application.

You are free to choose a project idea based on your interests. It could be related to education, tourism, cultural heritage, product visualization, entertainment, or another suitable area.

The main goal is to develop a working AR experience that demonstrates what you have learned during the WebAR sessions.

2. What You Need to Develop

Create a WebAR experience based on a concept of your choice.

Your project must include at least one 3D model placed in an AR environment and at least one feature that allows the user to interact with the experience.

You may create your own 3D models, use AI-generated models, or download suitable assets from online libraries, provided you have permission to use them.

The completed project must be published online and tested on a smartphone.

3. Suggested Project Ideas

You can choose one of the following ideas or develop something different.

Option 01 – Cultural Heritage AR

Create an AR experience featuring a historical building, monument, or cultural object. Users should be able to place the model in their surroundings and explore it.

Option 02 – Product Visualization AR

Develop an experience that allows users to view a product, such as furniture or an electronic device, in their own environment. You could also include options to change the product's colour or appearance.

Option 03 – Educational AR

Create an educational AR experience using 3D models. For example, you could display the solar system, human anatomy, animals, or scientific equipment with additional information.

Option 04 – Interactive Character AR

Create an AR character that users can place in the real world. The character could perform an animation or react when the user taps a button.

You are welcome to use your own idea as long as it meets the assignment requirements.

4. Minimum Requirements

Your WebAR project must include the following:

1. **Project Concept:** A clear idea with a purpose and intended audience.
2. **WebAR Platform:** Use an online platform that supports web-based AR experiences.
3. **3D Model:** Include at least one suitable 3D model with proper materials and textures.
4. **AR Placement:** The model must appear in the real-world environment using surface placement or image tracking.
5. **User Interaction:** Include at least one interactive feature.
6. **User Instructions:** Provide simple instructions so users know how to use the experience.
7. **Online Access:** Publish the experience and make it accessible through a web link.
8. **Mobile Testing:** Test the published experience on a compatible smartphone.

Interactive Feature

Your project must include at least one of the following interactions, or another suitable interaction supported by your chosen platform:

- Tap a button to play an animation.
- Show information about the 3D object.
- Change the object's colour or material.
- Switch between different 3D models.
- Move, rotate, or resize the object.

You do not need to include all these features. One properly working interaction is enough to meet the minimum requirement.

5. Development Process

Step 01 – Choose Your Idea

Decide what you want to create. Think about who will use your experience and what they should be able to do.

Step 02 – Prepare Your 3D Models

Create or download the models required for your project. Make sure they have suitable textures, materials, and scale. Keep the models optimized so they can load smoothly on mobile devices.

Step 03 – Build Your WebAR Experience

Import your 3D models into the WebAR platform. Set up the AR placement, adjust the scene, and add your interactive feature.

Step 04 – Publish Your Project

Once the experience is working, publish it online. Generate a web link and QR code that can be used to open the experience.

Step 05 – Test on a Smartphone

Open the published experience on a smartphone and check that the model displays correctly, interactions work, and the overall experience is easy to use.

Fix any issues before submitting your work.

6. What You Need to Submit

Each student must submit the following items:

1. WebAR Project Link

Provide the published URL of your WebAR experience. The link must be accessible and working.

2. QR Code

Submit a QR code that opens your published WebAR experience when scanned.

3. Demonstration Video (1–3 Minutes)

Record a short video showing your WebAR experience running on a smartphone.

The video should clearly show the AR model placed in the real-world environment and demonstrate the interactive feature you have developed.

4. Project Documentation (2–3 Pages)

Prepare a short report covering:

- Project title and concept
- Purpose and target audience
- Software and online platforms used
- Details of the 3D models
- Interactive features developed
- Screenshots of the experience
- Testing results and any problems encountered

7. Important Instructions

- This is an individual assignment. Each student must submit their own project.
- Your project idea should be original. If you use downloaded assets, include their sources in your documentation.
- AI tools may be used to help create models or other project content, but you must mention them in your report.
- Your final submission must be a working WebAR experience, not just a 3D model or a video.
- Make sure the published link and QR code work before submitting.
- Your project should be easy to access and use on a smartphone.

8. Expected Outcome

By the end of this assignment, you should be able to prepare 3D content for AR, build a basic interactive WebAR experience, publish it online, and test it on a smartphone.

9. Grading Criteria

Assessment Criteria	What Will Be Assessed	Marks
Project Concept and Purpose	A clear idea, suitable purpose, and identified target audience.	**5**
3D Model Quality and Preparation	Suitable models, materials, textures, scale, and mobile optimization.	**10**
AR Placement and Functionality	Successful placement or image tracking, correct model display, and reliable operation.	**10**
User Interaction	At least one working, easy-to-use interaction that supports the project's purpose.	**10**
User Experience and Instructions	Clear instructions, readable interface elements, and ease of use on a smartphone.	**5**
Publishing and Mobile Testing	A working published link and QR code, with evidence of smartphone testing.	**5**
Demonstration Video and Documentation	A clear 1–3 minute demonstration and complete 2–3 page report, including asset sources and AI tools used.	**5**
Total		**50**